

Weekly Progress Report

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Domain: Python Development Internship

Date of Submission: 9 July 2026

Week Ending: 04

I. Overview

This was the final week of my internship project. During this week, I focused on completing all four Python projects and making them ready for final submission. Most of my time was spent on integrating the project modules, fixing bugs, improving the user interface, testing every feature, and organizing the project files for GitHub. I also prepared the final documentation and verified that each project was working properly before submission.

II. Achievements

1. Final Project Completion

This week I successfully completed the development of all four Python projects and prepared them for final submission.

Project 1: URL Shortener

- Improved the web interface for better user experience.
- Tested URL shortening and redirection with different websites.
- Verified REST API endpoints using Postman.
- Improved database performance and click counter accuracy.
- Prepared the project for GitHub upload.

Project 2: File Organizer

- Optimized file sorting performance.
- Added better duplicate file handling.
- Verified custom category configuration.
- Tested the application with different folders containing documents, images, videos and compressed files.
- Fixed minor bugs during testing.

Project 3: Password Manager

- Verified password encryption and decryption process.
- Tested master password authentication.

- Improved password generation functionality.
- Checked backup and restore features.
- Fixed validation issues while storing passwords.

Project 4: Quiz Game

- Added more quiz questions.
- Improved score calculation.
- Verified timer functionality.
- Fixed small bugs in question selection.
- Tested the complete quiz flow from start to finish.

2. Documentation & GitHub

- Organized all project files into separate folders.
- Prepared README files for every project.
- Created requirements.txt files.
- Uploaded all project source code to GitHub.
- Prepared final documentation for submission.

3. Testing

This week I performed complete testing of all four projects.

Testing included:

- Functional Testing
- Input Validation
- Error Handling
- Database Testing
- API Testing
- Performance Testing

All projects worked successfully after fixing minor issues.

III. Challenges & Hurdles

GitHub Repository Management

Challenge:

Initially, organizing all project files and uploading them to GitHub in a proper folder structure was confusing.

Resolution:

I learned how to create repositories, organize folders, commit changes, and upload projects using GitHub.

Final Bug Fixing**Challenge:**

While performing final testing, I found a few small issues related to user input validation and database records.

Resolution:

I debugged the code carefully and corrected the validation logic. After testing again, the projects worked correctly.

Documentation Preparation**Challenge:**

Preparing documentation with installation steps, screenshots, and project descriptions required extra attention.

Resolution:

I created structured documentation and reviewed it before final submission.

IV. Lessons Learned

During this internship, I learned much more than just writing Python code.

Some important lessons I learned are:

- How to build complete Python applications from scratch.
- Importance of writing clean and organized code.
- Basic web development using Flask.
- Working with SQLite databases.
- Password encryption and application security.
- File handling and automation using Python.
- REST API development and testing.
- Debugging techniques and problem solving.
- Project documentation and GitHub version control.

This internship greatly improved both my programming skills and confidence in building real-world applications.

V. Learning Resources

- Python Official Documentation
- Flask Documentation
- SQLite Documentation
- Cryptography Library Documentation
- Stack Overflow
- GitHub Documentation
- YouTube Tutorials
- ChatGPT

VI. Final Outcome

By the end of this internship, I successfully completed the following Python projects:

- URL Shortener
- File Organizer
- Password Manager
- Quiz Game

All projects were tested successfully, documented properly, and uploaded to GitHub for final submission.

VII. Additional Comments

This internship was a great learning experience for me. I spent approximately **4–5 hours daily** learning, coding, debugging, and improving my projects. Every week I gained new knowledge, from basic Python programming to building complete applications with databases, APIs, and security features.

Completing all four projects has increased my confidence in Python development, and I look forward to applying these skills in future real-world projects and internships.